INSTRUCTOR LIVE TROUBLESHOOTING & Q&A; PLAYBOOK
Delivered by Tryolabs • Powered by Google Antigravity & Gemini Enterprise

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TRYOLABS • INSTRUCTOR LIVE TROUBLESHOOTING & ATTENDEE Q&A PLAYBOOK

What Could Go Wrong During the Labs: 16 Common Attendee Questions & Exact Fixes

Keep this quick-reference playbook on your podium during the 4-hour Google Antigravity workshop (Telecom & Retail tracks). Every question below represents a realistic failure mode an attendee might run into during Labs 01–04, paired with the exact root cause, what to say to the attendee, and the copy-paste terminal or code fix.

[Environment & Setup (All Labs)] Q1: 'When I run ./labs/lab-01-.../install.sh, I get python3: command not found, No module named venv, or an old Python version error.'

Root Cause: Attendee laptop is missing Python 3.10+ on PATH or lacks python3-venv on Ubuntu/Debian/WSL.

Instructor Answer & Exact Fix:
1. Check if **uv** is installed (**`uv --version`**). If so, run **`uv venv .venv --python 3.11 && source .venv/bin/activate && uv pip install -r requirements.txt`**.
2. On macOS with Homebrew: **`brew install python@3.11`**.
3. On Windows: Run PowerShell script **`.\labs\lab-01-...\install.ps1`** instead of **`.sh`**.
4. Always ensure commands are run using **`.venv/bin/python3`** and **`.venv/bin/pytest`**.

[Environment & Setup (All Labs)] Q2: 'When I start the FastAPI dashboard (uvicorn src.main:app --port 8000), it crashes with [Errno 48] Address already in use (macOS/Linux) or WinError 10048 (Windows).'

Root Cause: Another process or a previous terminal tab is already listening on port 8000 (or port 8080 for **`adk web`**).

Instructor Answer & Exact Fix:
• On macOS/Linux, free the port immediately: **`lsof -ti:8000 | xargs kill -9`** (or for port 8080: **`lsof -ti:8080 | xargs kill -9`**).
• Alternatively, run on a different port: **`.venv/bin/uvicorn src.main:app --reload --port 8005`**.

[Environment & Setup (All Labs)] Q3: 'I accidentally corrupted my SQLite database (data/*.db) or deleted part of a service file while experimenting. How do I recover quickly?'

Root Cause: Attendee ran a destructive query or bad edit during exploration.

Instructor Answer & Exact Fix:
• Re-run the current lab's installer: **`./labs/lab-0X-.../install.sh`**.
• Explain that every **`install.sh`** script is **self-healing and idempotent**: it automatically restores the clean starter state for the current lab, applies verified solutions for all prior labs, and re-seeds the SQLite database (**`python3 data/seed_telecom_data.py`** or **`python3 data/seed_retail_data.py`**) in 2 seconds!

[Environment & Setup (All Labs)] Q4: 'I didn't finish Lab 1 in time and now the class is moving to Lab 2. Am I stuck?'

Root Cause: Normal pacing variance across attendees in a hands-on coding workshop.

Instructor Answer & Exact Fix:
• Reassure the attendee: 'Not at all! Just run **`./labs/lab-02-rules-and-skills/install.sh`**.'
• Every lab's **`install.sh`** automatically copies the reference solution from prior labs into **`src/`** before setting up the new lab. And if anyone wants to see the full solution to the current lab, they can run **`./labs/lab-0X-.../apply_solution.sh`**.

[Antigravity & Gemini Enterprise Auth] Q5: 'Antigravity gives a 403 PermissionDenied error or says cloudaicompanion.googleapis.com is not enabled for my account.'

Root Cause: The attendee's corporate Google identity is either logged into a personal Gmail profile or missing the IAM role binding on the workshop GCP project.

Instructor Answer & Exact Fix:
1. Verify the attendee is signed into Antigravity using their corporate email associated with the workshop.
2. In the Instructor GCP Console -> **IAM & Admin**, grant the attendee's email two roles: **`Gemini for Google Cloud User`** (**`roles/cloudaicompanion.user`**) and **`Service Usage Consumer`** (**`roles/serviceusage.serviceUsageConsumer`**).
3. Ensure **`cloudaicompanion.googleapis.com`** is enabled in **APIs & Services**.

[Lab 01 — Telecom Math & Logic] Q6 (Telecom): 'Antigravity updated cdr_rating.py, but pytest still fails on the CDR charge amount by ~7.37% or on NR_5G roaming!'

Root Cause: Two classic Telecom BSS pitfalls in **`src/services/cdr_rating.py`**: (1) Dividing total bytes by decimal GB (**`10**9 = 1,000,000,000`**) instead of 3GPP binary GiB (**`2**30 = 1024**3 = 1,073,741,824`**); or (2) Only applying **`roaming_markup`** when **`rat_type == 'LTE'`** and ignoring **`'NR_5G'`** visiting PLMN roaming.

Instructor Answer & Exact Fix:
• Show the exact formula using **`Decimal`**: **`volume_gib = Decimal(uplink_bytes + downlink_bytes) / Decimal(1024 ** 3)`**.
• Apply **`roaming_markup`** whenever **`visiting_plmn != home_plmn`** across both **`'LTE'`** and **`'NR_5G'`**, and quantize to **`Decimal('0.01')`** using **`ROUND_HALF_UP`** only at the end!

[Lab 01 — Telecom Math & Logic] Q7 (Telecom): 'Why is sla_monitor.py either including MAINTENANCE cells or missing active URLLC latency breaches?'

Root Cause: In **`src/services/sla_monitor.py`**, the starter query omits **`WHERE status = 'ACTIVE'`** and uses **<** instead of **>** (**`worst_latency < committed_latency`**).

Instructor Answer & Exact Fix:
• Ensure the SQL query filters **`WHERE c.status = 'ACTIVE'`** so maintenance windows are excluded.
• Trigger a breach when **`worst_latency > committed_latency`** or **`worst_jitter > committed_jitter`**, and compute **`hourly_penalty_exposure_usd`**.

[Lab 01 — Retail Math & Logic] Q8 (Retail): 'In promo_pricing.py, I enforced a margin floor using unit_cost * (1 + floor_pct), so why is pytest saying my minimum price is wrong?'

Root Cause: Classic confusion between **Markup** (**`cost * (1 + pct)`**) and **Gross Margin** (**`(price - cost) / price = pct`**).

Instructor Answer & Exact Fix:
• Write both formulas on the screen—attendees love this clarification:
- **Markup** divides profit by cost: **`price = cost * 1.15`** (if cost is $85, 15% markup gives $97.75, which is only a 13.04% gross margin!).
- **Gross Margin** divides profit by net selling price: **`(price - cost) / price >= floor_pct`**, which algebraically rearranges to **`min_price = unit_cost / (Decimal('1') - floor_pct)`** (if cost is $85 and floor is 15%, **`$85 / 0.85 = $100.00`**).
• Also confirm **`stackable`** promos are applied multiplicatively (**`price = price * (1 - d1) * (1 - d2)`**), never added (**`d1 + d2`**).

[Lab 01 — Retail Math & Logic] Q9 (Retail): 'Why does my GS1 GTIN-14 barcode validator reject valid 14-digit barcodes in inventory_atp.py?'

Root Cause: Reversing the 3-and-1 multiplier weights in the GS1 Modulo-10 algorithm.

Instructor Answer & Exact Fix:
• For a 14-digit **`GTIN-14`**, the first 13 digits (**`digits[:13]`**) are the payload and the 14th digit (**`digits[13]`**) is the check digit.
• Moving from left to right across the first 13 digits (index **`0`** to **`12`**), even indices (**`0, 2, 4, ... 12`**) are multiplied by **`3`**, and odd indices (**`1, 3, 5, ... 11`**) are multiplied by **`1`**. Then **`check_digit = (10 - (total % 10)) % 10`**.

[Lab 02 — Rules & Skills] Q10: 'I created SKILL.md inside .agents/skills/, why doesn't Antigravity discover or activate my skill when I prompt it?'

Root Cause: Progressive disclosure discovery relies strictly on the YAML frontmatter at the top of **`SKILL.md`** and the exact directory path.

Instructor Answer & Exact Fix:
Check these 3 items with the attendee:
1. **Exact Path & Case:** Must be **`.agents/skills/<skill-name>/SKILL.md`** (**`SKILL.md`** in ALL CAPS, inside **`.agents/skills/`**).
2. **Valid YAML Frontmatter:** Line 1 of **`SKILL.md`** must be **`---`**, followed by **`name:`** and **`description:`**, closed by **`---`**.
3. **Trigger Keywords in Description:** Antigravity only loads **`name`** and **`description`** initially. Make sure **`description`** clearly states when to activate (e.g., *'Use when generating an executive SLA revenue risk audit...'*).

[Lab 02 — Rules & Skills] Q11: 'When Antigravity runs my skill script (.agents/skills/.../scripts/run_audit.py), it fails with ModuleNotFoundError: No module named src.'

Root Cause: Python does not automatically add the workspace root to **`sys.path`** when executing a script nested four directories deep inside **`.agents/skills/<name>/scripts/`**.

Instructor Answer & Exact Fix:
• Either run the script with **`PYTHONPATH=. .venv/bin/python3 .agents/skills/.../scripts/run_audit.py`**, OR
• Add **`sys.path.insert(0, str(Path(__file__).resolve().parents[4]))`** at the top of **`run_audit.py`** before importing from **`src.*`**.

[Lab 03 — FastMCP & Hooks] Q12: 'My FastMCP server works when I import it in Python, but Antigravity says MCP connection failed or invalid JSON-RPC frame!'

Root Cause: The #1 cause of **`stdio`** MCP bugs: someone added a **`print('Connected to DB...')`** statement that writes to **`stdout`**!

Instructor Answer & Exact Fix:
• Explain how **`stdio`** MCP transport works: Antigravity spawns **`mcp_server/*_mcp.py`** as a subprocess and communicates over **`stdin`** and **`stdout`** using strict JSON-RPC 2.0 messages.
• If any code prints plain text to **`stdout`**, it corrupts the JSON-RPC stream! Any debug logs inside an MCP server MUST use **`print(..., file=sys.stderr)`** or Python's **`logging`** module configured to **`stderr`**.
• Also verify **`.agents/mcp_config.json`** uses **`.venv/bin/python3`** (not system **`python3`** which might lack the **`mcp`** package).

[Lab 03 — FastMCP & Hooks] Q13: 'My PreToolUse hook script exits with code 1 on a violation, why didn't Antigravity treat it as a deterministic policy block?'

Root Cause: In Antigravity's hook specification, exit code **`1`** indicates a non-fatal script crash/warning, whereas **exit code 2** is the explicit signal for a deterministic policy block!

Instructor Answer & Exact Fix:
• Review the Antigravity Hook Exit Code Contract:
- **`sys.exit(0)`** -> Action is compliant; proceed with tool execution.
- **`sys.exit(2)`** -> **Deterministic Policy Block!** Antigravity aborts the tool call immediately and feeds the hook's **`stderr`** explanation back to the agent.
• Also ensure the script reads the tool payload via **`payload = json.load(sys.stdin)`** and writes an audit entry to **`logs/*.jsonl`** before calling **`sys.exit(2)`**.

[Lab 04 — ADK, adk web, A2A & Terraform] Q14: 'When I launch ./scripts/run_adk_web.sh (http://localhost:8080) and send a chat message in the ADK Dev UI, it asks for GOOGLE_API_KEY or Vertex AI credentials. How does that work?'

Root Cause: `pytest` and the FastAPI A2A endpoint include a deterministic offline engine so local tests always pass without external API keys, whereas `adk web` invokes the live Gemini LLM inside `google.adk.agents.LlmAgent`.

Instructor Answer & Exact Fix:
• To chat live with Gemini inside `adk web`, configure either:
- **Option A (Vertex AI / Gemini Enterprise GCP Project):**

```
export GOOGLE_GENAI_USE_VERTEXAI=TRUE
export GOOGLE_CLOUD_PROJECT=<your-workshop-gcp-project-id>
export GOOGLE_CLOUD_LOCATION=us-central1
gcloud auth application-default login
```

- **Option B (Gemini API Key):**

```
export GOOGLE_API_KEY=<your-api-key>
```

• Then restart `./scripts/run_adk_web.sh` and open `http://localhost:8080`!

[Lab 04 — ADK, adk web, A2A & Terraform] Q15: 'In the ADK Dev UI (http://localhost:8080), the agent dropdown menu is empty or doesn't list telecom_nexus / retail_pulse!'

Root Cause: `adk web` expects a specific directory convention (`adk_agents/<agent_name>/__init__.py` and `agent.py` exporting a variable named `root_agent`).

Instructor Answer & Exact Fix:
• Always launch `adk web` via `./scripts/run_adk_web.sh` (which runs `.venv/bin/adk web adk_agents --port 8080` from the repository root).
• Verify that `adk_agents/<agent_name>/agent.py` defines `root_agent = LlmAgent(...)` and `adk_agents/<agent_name>/__init__.py` contains `from . import agent`.

[Lab 04 — ADK, adk web, A2A & Terraform] Q16: 'When I run ./deploy/deploy_gcp.sh to test the Terraform deployment, I don't have Docker Desktop running or GCP project owner permissions on my laptop.'

Root Cause: Attendee ran `./deploy/deploy_gcp.sh` without `--dry-run` on a laptop without Docker or active `gcloud` credentials.

Instructor Answer & Exact Fix:
• For local verification on attendee laptops, run `./deploy/deploy_gcp.sh --dry-run` (or `./labs/lab-04-multi-agent-capstone/deploy_gcp.sh --dry-run`).
• Explain that `--dry-run` validates all Terraform HCL files (`infra/terraform/main.tf`, `variables.tf`, `outputs.tf`), Dockerfile, and Gemini Enterprise registration payloads 100% locally without requiring Docker daemon or live GCP privileges!